

计算机体系结构

1. 课程简介

李建华

计算机与信息学院

The contents of the slides are adapted from CA course of wisc, princeton, mit, berkeley, edinburgh, and eth.
The uses of the slides of this course are for educational purposes only and should be used only in conjunction with the textbook. Derivatives of the slides must acknowledge the copyright notices of this and the originals.

主页: http://faculty.hfut.edu.cn/jhli_hfut/zh_CN/index.htm

个人简介

李建华, 博士, 副研究员, 主要从事计算机系统方向的教学和科研工作, 主要研究方向为人工智能与数字系统、片上存储和片上网络的交叉研究。近年来在领域内核心会议和期刊上, 如IEEE TC, IEEE TPDS, ACM TODAES, 电子学报、计算机学报等, 发表学术论文40多篇。获2015年香江学者奖, 2024年IEEE Andrew P. Sage Best Transaction Paper Award。

研究方向:

>> AI与计算机系统交叉: 如基于强化学习的缓存预取策略、基于强化学习的片上网络路由算法设计等;

>> FPGA开发及应用: 如基于FPGA的AI加速器设计、基于FPGA的高并行系统开发 (如芯片测试仪器的控制系统开发) 。

>> 欢迎: **热爱计算机系统、喜欢折腾底层、想提升动手能力** (low-level coding and debugging) 的同学联系: [jhli @ hfut.edu.cn](mailto:jhli@hfut.edu.cn)

>> 喜欢的系统名言: (1) Get your hand dirty! (2) KISS: Keep it simple, stupid! (3) Talk is cheap, show me the code!

教育背景:

>> 2007年获得安庆师范大学计算机科学与技术专业学士学位;

>> 2013年获得中国科学技术大学软件与理论专业博士学位;

>> 2013年8月起, 在合肥工业大学计算机与信息学院工作;

>> 2016-2018, 香港城市大学, Hong Kong, 香江学者。

讲授课程:

>> 本科生: 计算机体系结构、系统硬件综合设计;

学科竞赛方面, 指导本科生获得安徽省机器人大赛CPU设计赛道一等奖10余项, 获得龙芯杯个人赛二等奖2项。

>> 研究生: 高级计算机体系结构、深入理解计算机系统。

微信: [rallylee_micro](#) 邮箱: jhli@hfut.edu.cn

计算与抽象

计算机体系结构
(扩展视图)



有没有利用物理特性
直接可以做应用来解决
问题的例子?

计算机体系结构
(狭窄视图)

抽象的作用非常重要!

基于抽象来解决计算问题

算法

能确保结束的一个分步骤的过程，每个过程均能精确地进行描述，且可以被计算机进行处理。

- 有限性
- 确定性
- 可计算性

针对同一个问题，有很多不同的算法。

微架构

是ISA具体的实现；同一个ISA可以有不同的微架构实现。

应用(问题)

算法

编程语言

系统软件

(VM, OS, MM, Compiler)

ISA (体系结构)

微架构

逻辑电路

器件

电子学

ISA

(Instruction Set Architecture)

软件和硬件之间的接口或者契约。

程序员希望硬件能够具备的功能是什么？

数字逻辑电路

是构建微架构的基础部件 (例如：逻辑门、MUX)。

影响计算机性能提升的因素

- 计算机性能的提升依赖于:
 - ① 半导体技术工艺的进步 (**PUSH**)
 - Feature size
 - Clock speed
 - ② 计算机体系结构和系统的创新 (**FEEDBACK**)
 - 得益于先进的编译器, UNIX
 - 产生了RISC架构 (当前的主导架构)
 - ③ 核心应用的拉动作用 (**PULL**)
 - 吸引资源投入, 如AI (**Meta的招人方式**)
- 前二者的结合, 导致:
 - 轻量级计算机的普及
 - 各类高生产力的编程语言的出现

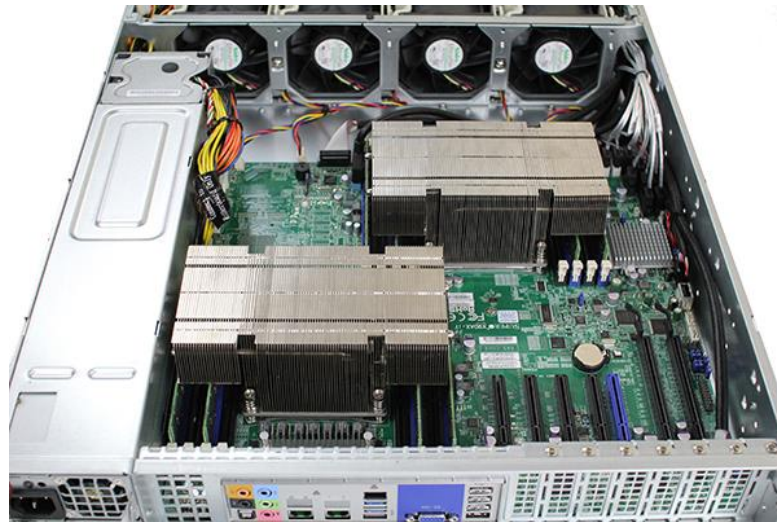
计算机的分类

- 个人移动设备 (PMD)
 - 例如：智能手机、平板，强调 能效、价格和实时性
- 桌面计算机 (Desktop)
 - 强调性价比



计算机的分类

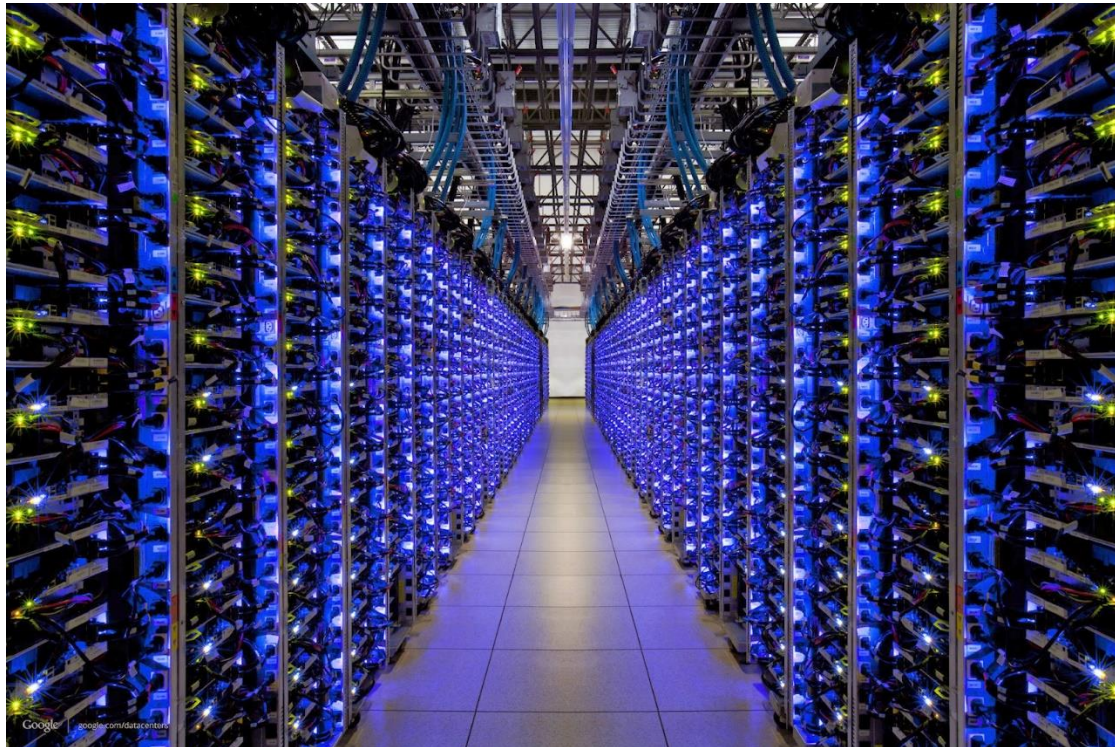
- 服务器 (Server)
 - 强调 可用性、可扩展性，和吞吐率



<https://www.servethehome.com/supermicro-hyper-speed-server-internal-components/>

计算机的分类

- 集群/数据仓库级计算机 (WSC)
 - 用于 “Software as a Service (SaaS)” 等
 - 强调 可用性 和 性价比



计算机的分类

- 嵌入式计算机 (Embedded)
 - 因为量大类多，所以强调价格，如汽车电子



前沿的计算机：TPU

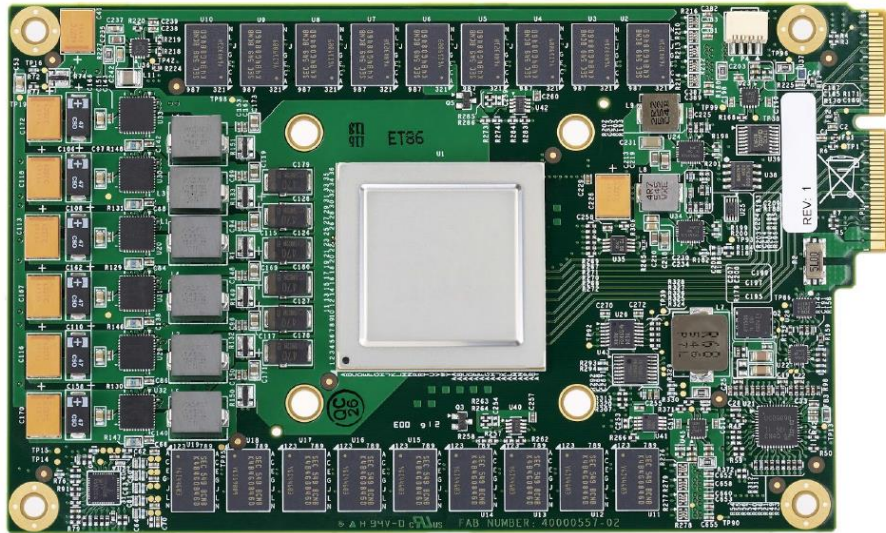


Figure 3. TPU Printed Circuit Board. It can be inserted in the slot for an SATA disk in a server, but the card uses PCIe Gen3 x16.

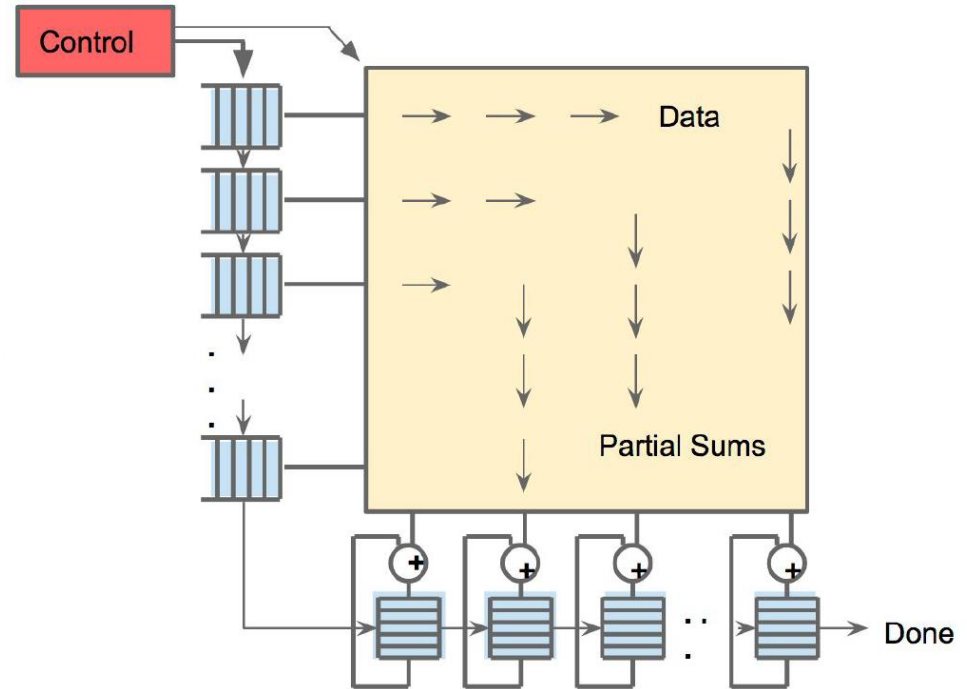
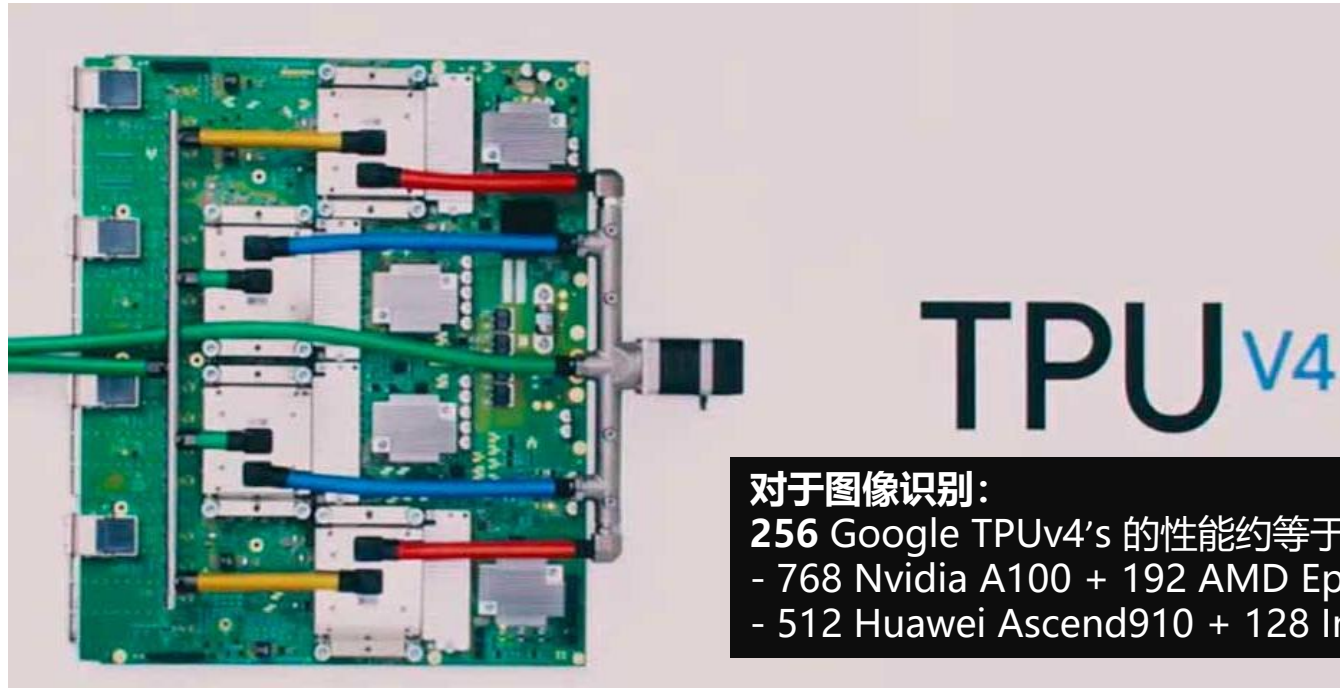


Figure 4. Systolic data flow of the Matrix Multiply Unit. Software has the illusion that each 256B input is read at once, and they instantly update one location of each of 256 accumulator RAMs.

Jouppi et al., “[In-Datcenter Performance Analysis of a Tensor Processing Unit](#)”, ISCA 2017.

前沿的计算机：TPU v4



对于图像识别：

256 Google TPUv4's 的性能约等于

- 768 Nvidia A100 + 192 AMD Epyc 7742 CPU
- 512 Huawei Ascend910 + 128 Intel Xeon 8168

新的机器学习应用 (vs. TPU3):

- 计算机视觉
- 自然语言处理
- 推荐系统
- 能玩围棋的强化学习

250 TFLOPS per chip in 2021
vs 90 TFLOPS in TPU3

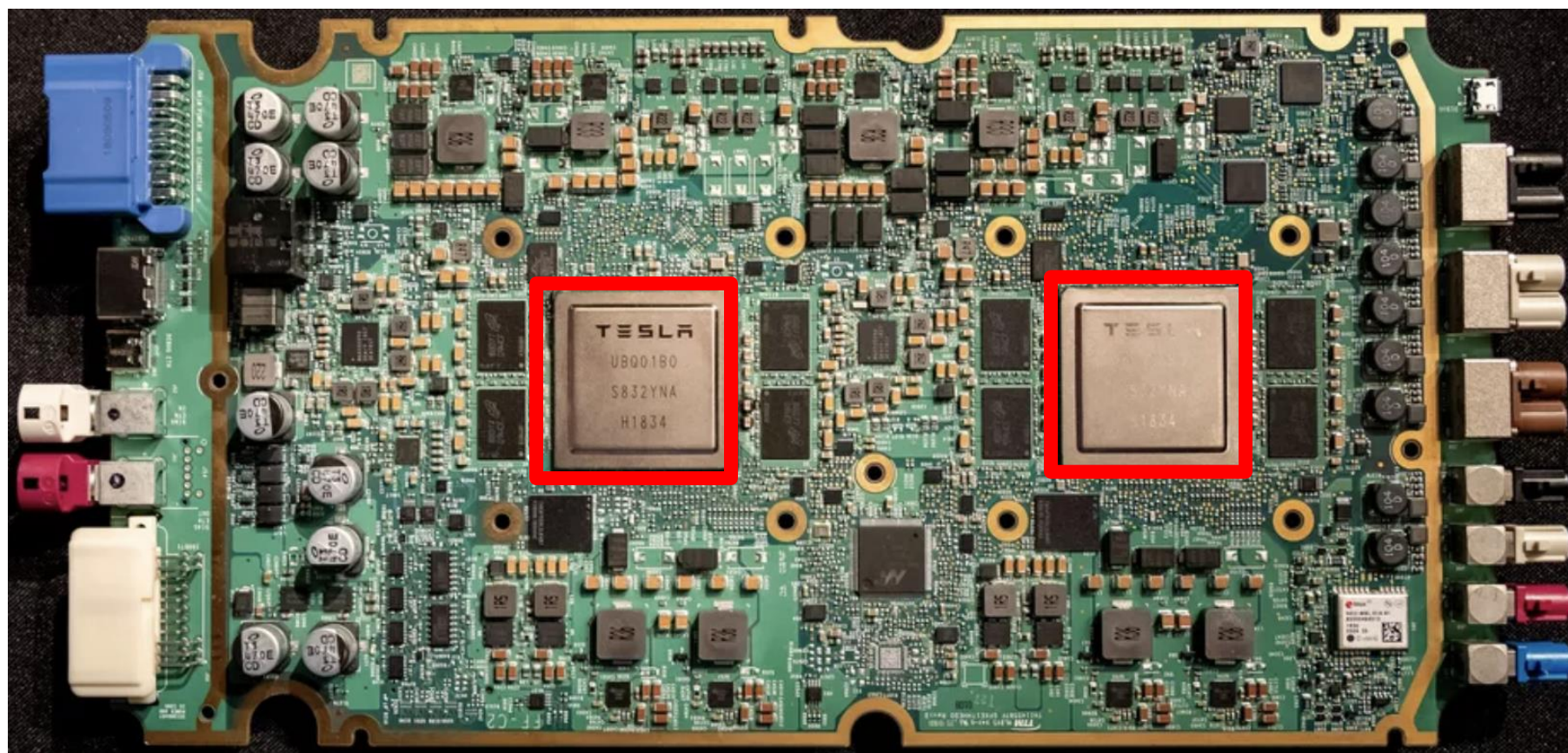


1 ExaFLOPS per board

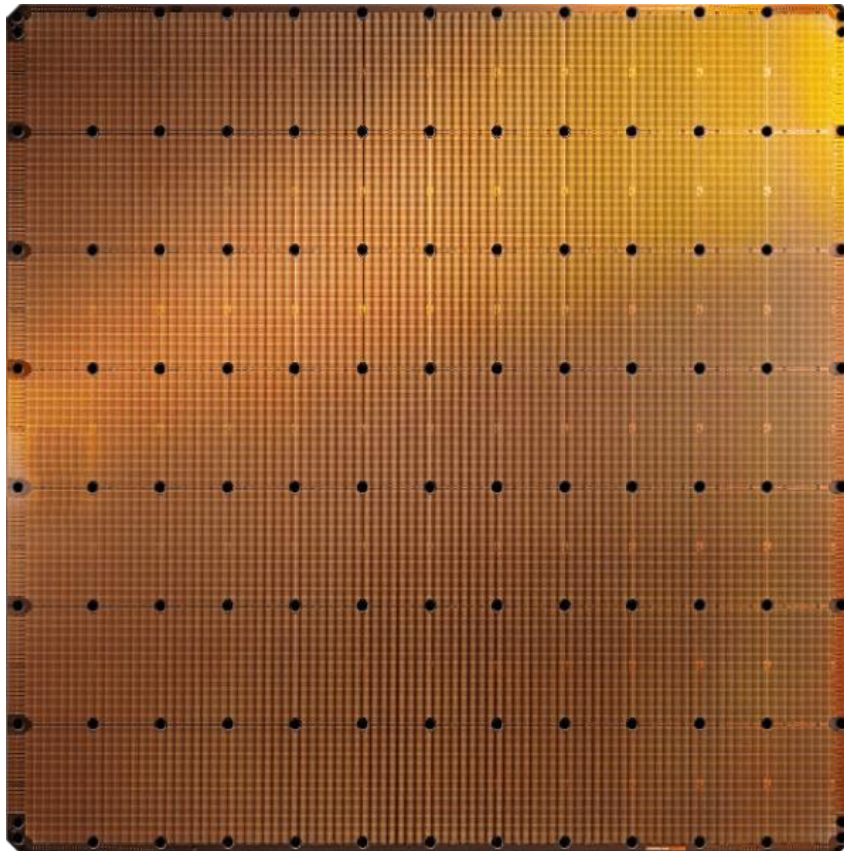
<https://spectrum.ieee.org/tech-talk/computing/hardware/heres-how-googles-tpu-v4-ai-chip-stacked-up-in-training-tests>

前沿的计算机：ML加速器

- ML加速器: 260 mm², 60亿晶体管, 600 GFLOPS GPU, 12 ARM 2.2 GHz CPUs.
- 为了安全, 使用冗余芯片技术。



前沿的计算机：WSE-2



Cerebras WSE-2

2.6 Trillion transistors, 46,225 mm²

- 最大的ML加速器芯片 (2021)
- 850,000 个核芯



Largest GPU

54.2 Billion transistors

826 mm²

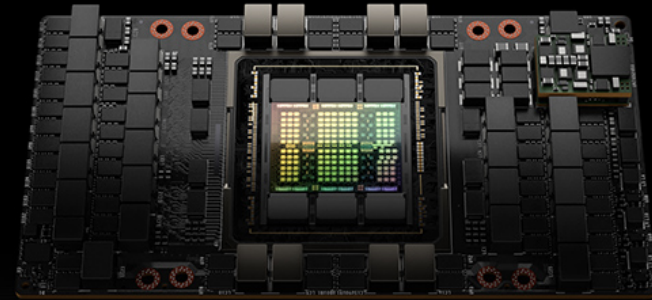
NVIDIA Ampere GA100

前沿的计算机：H100

NVIDIA H100 Tensor Core GPU

Unprecedented performance, scalability, and security for every data center.

Learn More

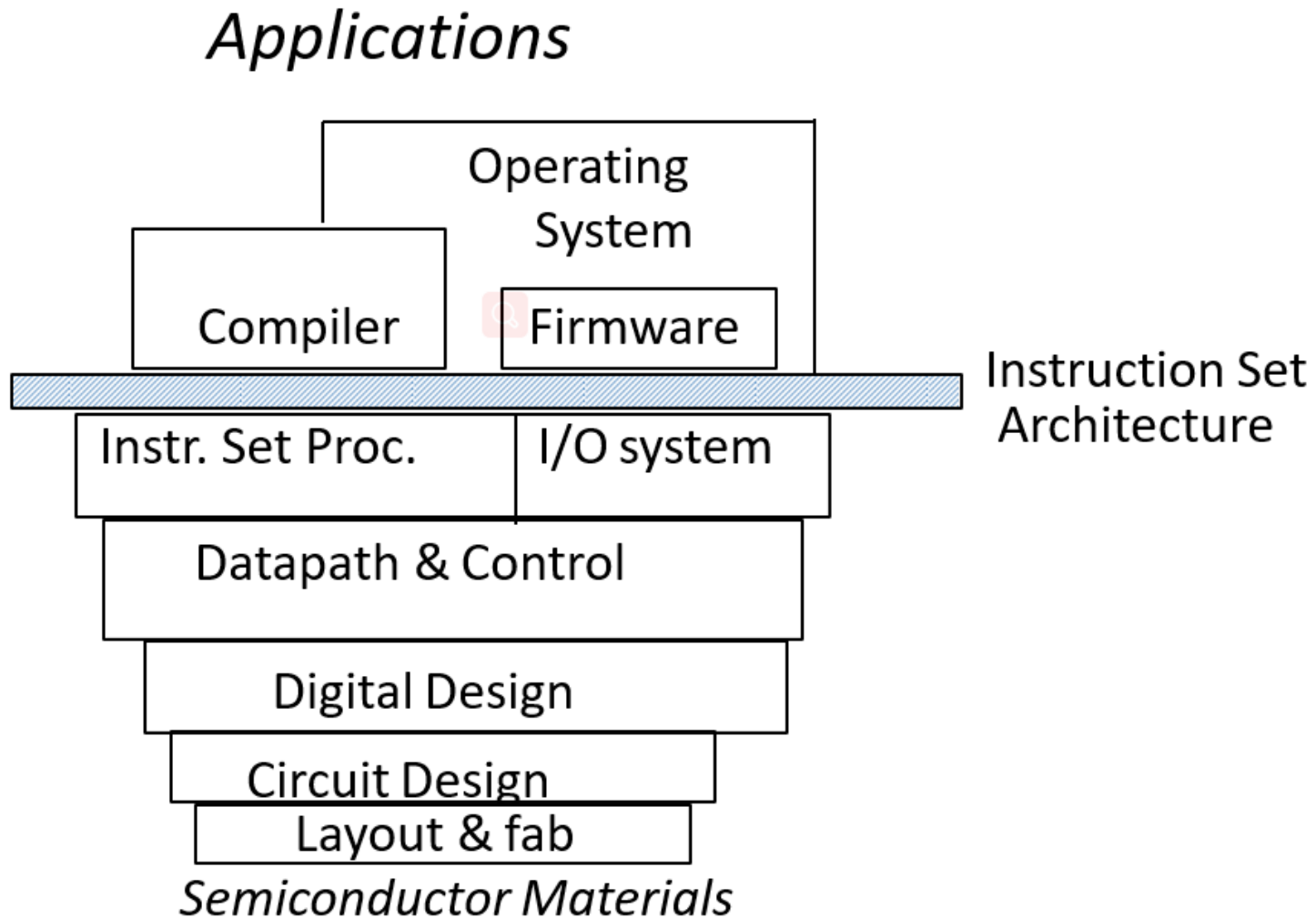


An Order-of-Magnitude Leap for Accelerated Computing

Tap into unprecedented performance, scalability, and security for every workload with the NVIDIA® H100 Tensor Core GPU. With the NVIDIA NVLink® Switch System, up to 256 H100 GPUs can be connected to accelerate exascale workloads. The GPU also includes a dedicated Transformer Engine to solve trillion-parameter language models. The H100's combined technology innovations can speed up large language models (LLMs) by an incredible 30X over the previous generation to deliver industry-leading conversational AI.

<https://www.nvidia.com/en-us/data-center/h100/>

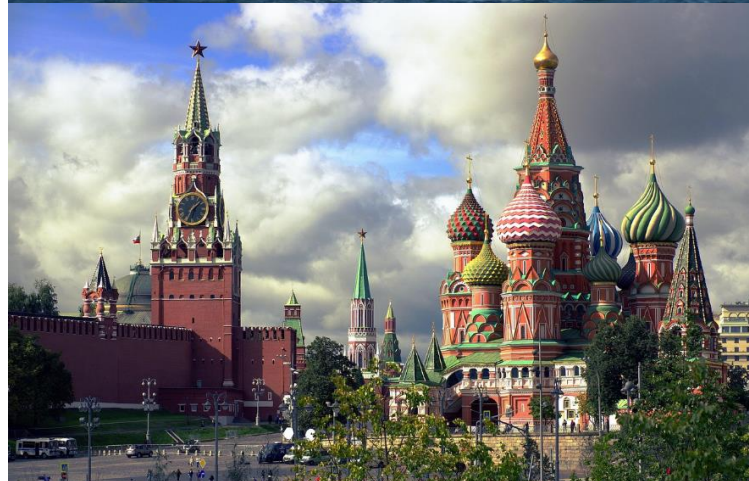
不同计算平台，共同的抽象特性



什么是计算机体系结构？

- The **science** and **art** of [designing](#), [selecting](#), and [interconnecting](#) hardware components and [designing](#) the hardware/software interface to create a **computing system** that **meets functional, performance, energy consumption, cost**, and other specific **goals**.
- 当前，计算机体系结构的发展趋势是什么？
 - Domain-Specific X (Accelerator)
 - 课后收看：ISCA 2018 [Turing Award 演讲](#)

与建筑学的类比



这些建筑有何共同的特性？

问题？

- 为什么要学习计算机体系结构？考虑一下，请说说你的想法。
 - 有些学校已经不开设计算机体系结构
 - 有些学校的计算机专业没有硬件类课程

我的回答：

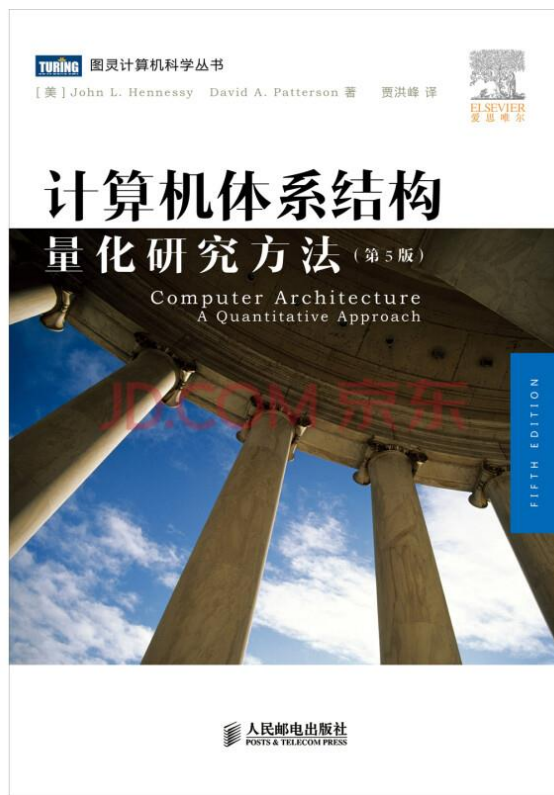
- **设计**更好的系统，获得较高的薪水。
 - 使得计算机更快、更小、更可靠...
 - Intel/AMD/Nvidia/华为/大疆/中芯国际...
- **赋能**新的应用
 - 自动驾驶？
 - 具身智能？
- **研究**更好的解决方案
 - 软件的创新依赖计算机体系结构的创新和改变
 - 成为一个**更好**的程序员
 - Richard Stallman, Donald Knuth, Linus Torvalds, Jeff Dean, Sanjay Ghemawat ...

知名教授的回答：

- **To become a computer designer**
 - Alumni of this class could help design your computer
- **To learn what is *under the hood* of a computer**
 - Innate curiosity
 - To better understand when programs break
 - To write better code/applications
 - To write better system software (OS, compiler, etc.)
- **Because it is intellectually fascinating!**
 - CPUs are arguably the most complex highly-integrated man-made devices

教材信息

- 参考教材：
 - 《计算机体系结构：量化研究方法》，第5版或6版；
 - 教材为主，相关书籍、学术论文和技术文档为辅。



课程内容

模块 1

- **Instruction Set Architecture (ISA)**
- **Cache Memory and Virtual Memory**
- **Simple Pipelining and Hazards**

模块 2

- **Complex Pipelining**
- **Out of Order Execution**
- **Branch Prediction and Speculative Execution**

模块 3

- **Vector Processor, Multicore Processor**
- **Coherence and Consistency**
- **On-Chip Networks**

课程考核

- 考核方式：

- 课后作业 — 3次作业
- 实验上机 — 3次实验，实验报告
- 期末报告 — 阅读论文，写报告
- 期末考试 — 闭卷，有一定难度

代码	名称	占比
EM2	实验上机	30%
EM20	期末考试	40%
EM4	期末报告（ <u>务必重视</u> ）	15%
EM17	课后作业（ <u>课堂测试</u> ）	15%

- 亲自动手去实践课程中的理论知识
- 为后续《系统硬件综合设计》做准备
- 可参加：机器人大赛CPU设计，龙芯杯 ...

CPU设计获奖



2017-1



2018-1



2018-2



2019-1



2019-2



2020-1



2020-2



2020-3



2022-1



2022-3



2022-4



2023-1



2023-2



2023-4



龙芯杯-李建华



龙芯杯-赵凌霄

2024: 2个省一等奖, 2个省二等奖, 1个龙芯杯个人赛二等奖

8 Great Ideas in CA

1. Design for *Moore's Law*
2. Use *abstraction* to simplify design
3. Make the *common case fast*
4. Performance via *parallelism*
5. Performance via *pipelining*
6. Performance via *prediction*
7. *Hierarchy* of memories
8. Dependability via *redundancy*



学完这门课，再回来看这里的内容。

技术趋势

- 集成电路技术 (摩尔定律)
 - Transistor density: 35%/year
 - Die size: 10-20%/year
 - Integration overall: 40-55%/year
- DRAM容量: 25-40%/year (slowing)
 - 8 Gb (2014), 16 Gb (2019), 32 Gb (2024?)
- Flash容量: 50-60%/year
 - 8-10X cheaper/bit than DRAM
- 机械硬盘容量: 最近下降到5%/year
 - 进一步增加存储密度可能行不通, 有可能盘片会增加到7~9个
 - 相比Flash, 8-10X cheaper/bit
 - 相比DRAM, 200-300X cheaper/bit

Transistors 与 Wires

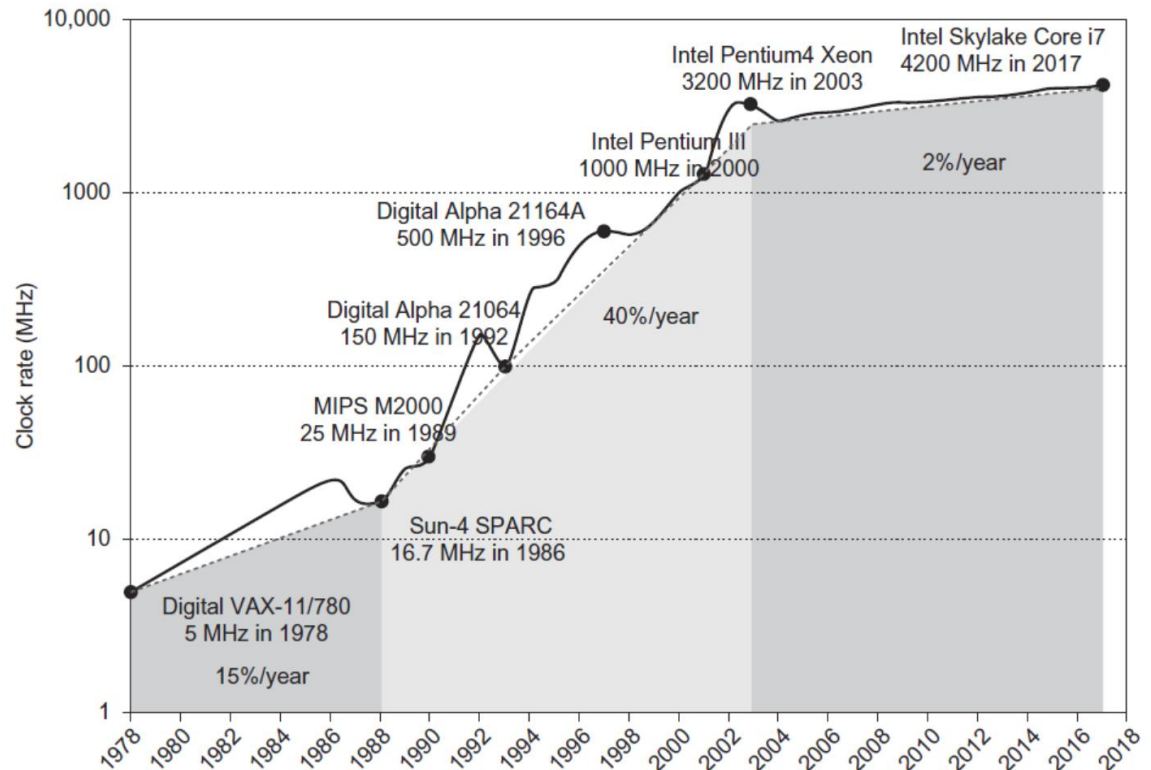
- 特征尺寸 (Feature size)
 - Transistor或Wire在 x 或者 y 维度的最小尺寸
 - 10 microns in 1971 to .011 microns in 2017
 - 当前的高性能芯片: 3~5nm
 - Transistor性能线性增加
 - Wire 延迟并没有随着特征尺寸的缩小而减少!
 - Transistor的集成密度呈平方地增加

动态能耗和功耗

- 动态能耗
 - Transistor在 0 -> 1 or 1 -> 0 之间切换而产生
 - 等于: $\frac{1}{2} \times \text{Capacitive load} \times \text{Voltage}^2$
- 动态功耗:
 - $\frac{1}{2} \times \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$ switched
- 所以, 降低 时钟频率 可以减少功耗, 而**不一定可以减少能耗。**

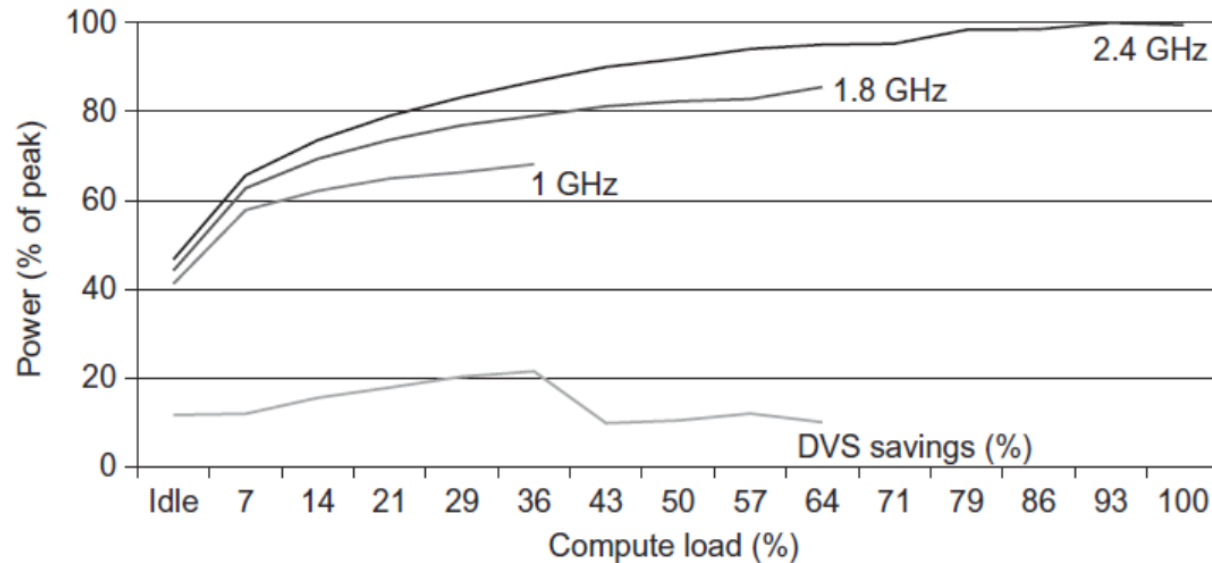
功耗问题

- Intel 80386 的功耗: $\sim 2\text{ W}$
- 3.3 GHz Intel Core i7 的功耗: 130 W
- 产生的热量需要从 $1.5 \times 1.5\text{ cm}^2$ 的芯片上散发掉
- 这是风冷技术可以处理的极限



如何降低功耗？

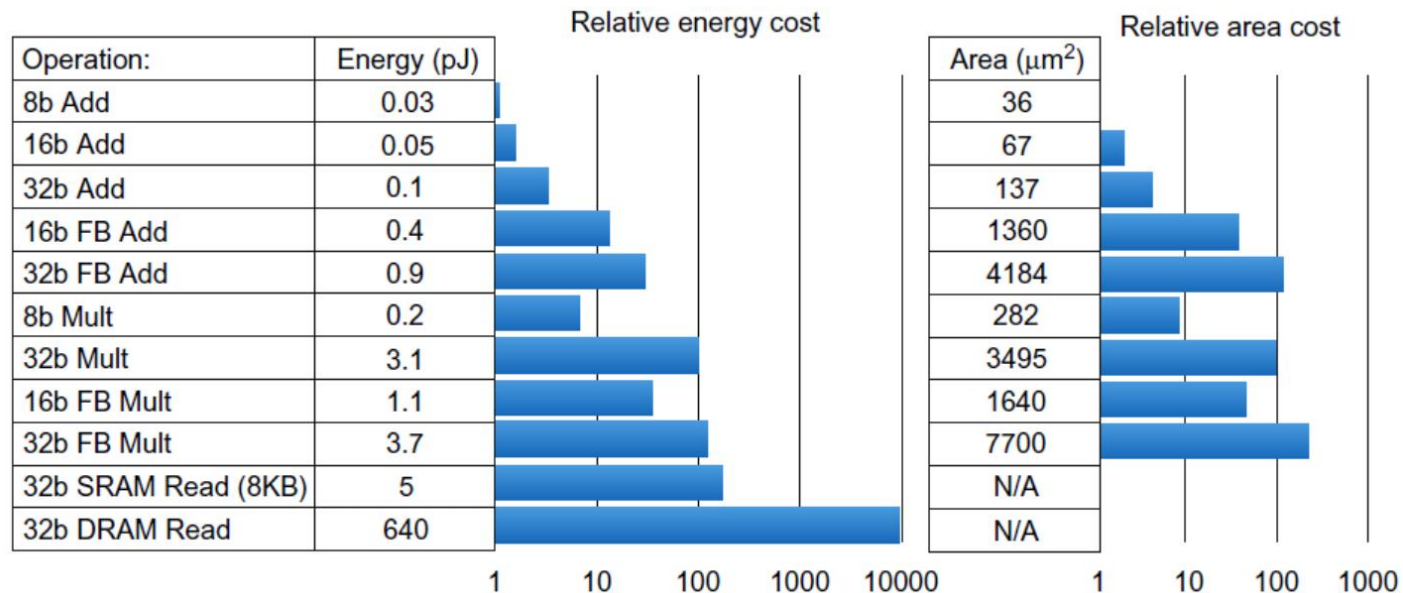
- 降低功耗的技术：
 - 啥也不做是最好的
 - Dynamic Voltage-Frequency Scaling



- DRAM, Disks引入低功耗状态
- 超频，动态关闭cores

静态功耗

- 静态功耗
 - 占据总功耗的：25-50%
 - $\text{Current}_{\text{static}} \times \text{Voltage}$
 - 随着Transistor数量增加而增加
 - 降低静态功耗的方法：power gating



Moore's Law (摩尔定律)

Cramming more components onto integrated circuits

With unit cost falling as the number of components per circuit rises, by 1975 economics may dictate squeezing as many as 65,000 components on a single silicon chip

By Gordon E. Moore

Director, Research and Development Laboratories, Fairchild Semiconductor division of Fairchild Camera and Instrument Corp.

The future of integrated electronics is the future of electronics itself. The advantages of integration will bring about a proliferation of electronics, pushing this science into many new areas.

Integrated circuits will lead to such wonders as home computers—or at least terminals connected to a central computer—automatic controls for automobiles, and personal portable communications equipment. The electronic wrist-watch needs only a display to be feasible today.

But the biggest potential lies in the production of large systems. In telephone communications, integrated circuits in digital filters will separate channels on multiplex equipment. Integrated circuits will also switch telephone circuits and perform data processing.

Computers will be more powerful, and will be organized in completely different ways. For example, memories built of integrated electronics may be distributed throughout the

machine instead of being concentrated in a central unit. In addition, the improved reliability made possible by integrated circuits will allow the construction of larger processing units. Machines similar to those in existence today will be built at lower costs and with faster turn-around.

Present and future

By integrated electronics, I mean all the various technologies which are referred to as microelectronics today as well as any additional ones that result in electronics functions supplied to the user as irreducible units. These technologies were first investigated in the late 1950's. The object was to miniaturize electronics equipment to include increasingly complex electronic functions in limited space with minimum weight. Several approaches evolved, including microassembly techniques for individual components, thin-film structures and semiconductor integrated circuits.

Each approach evolved rapidly and converged so that each borrowed techniques from another. Many researchers believe the way of the future to be a combination of the various approaches.

Electronics

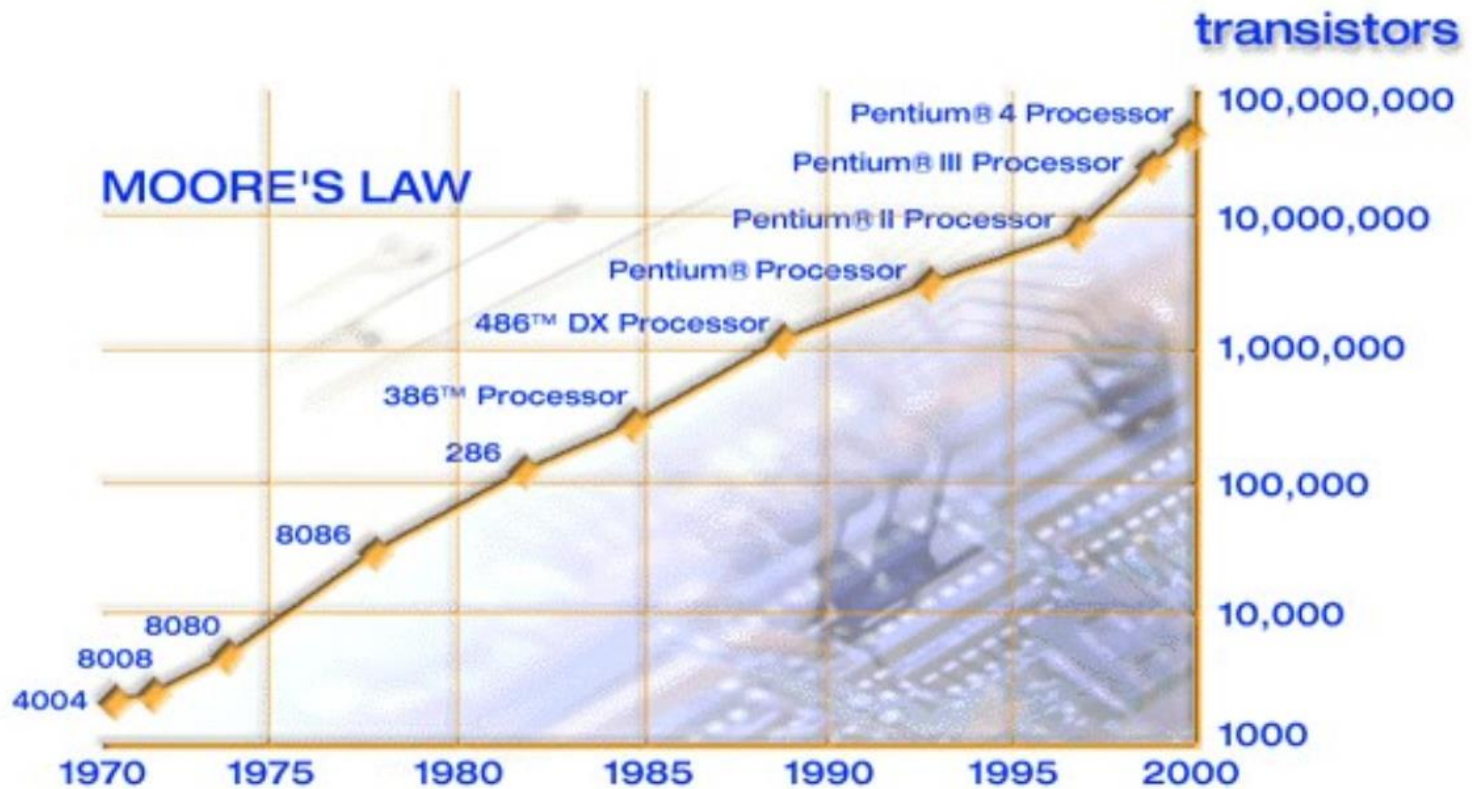
19 April 1965



The author

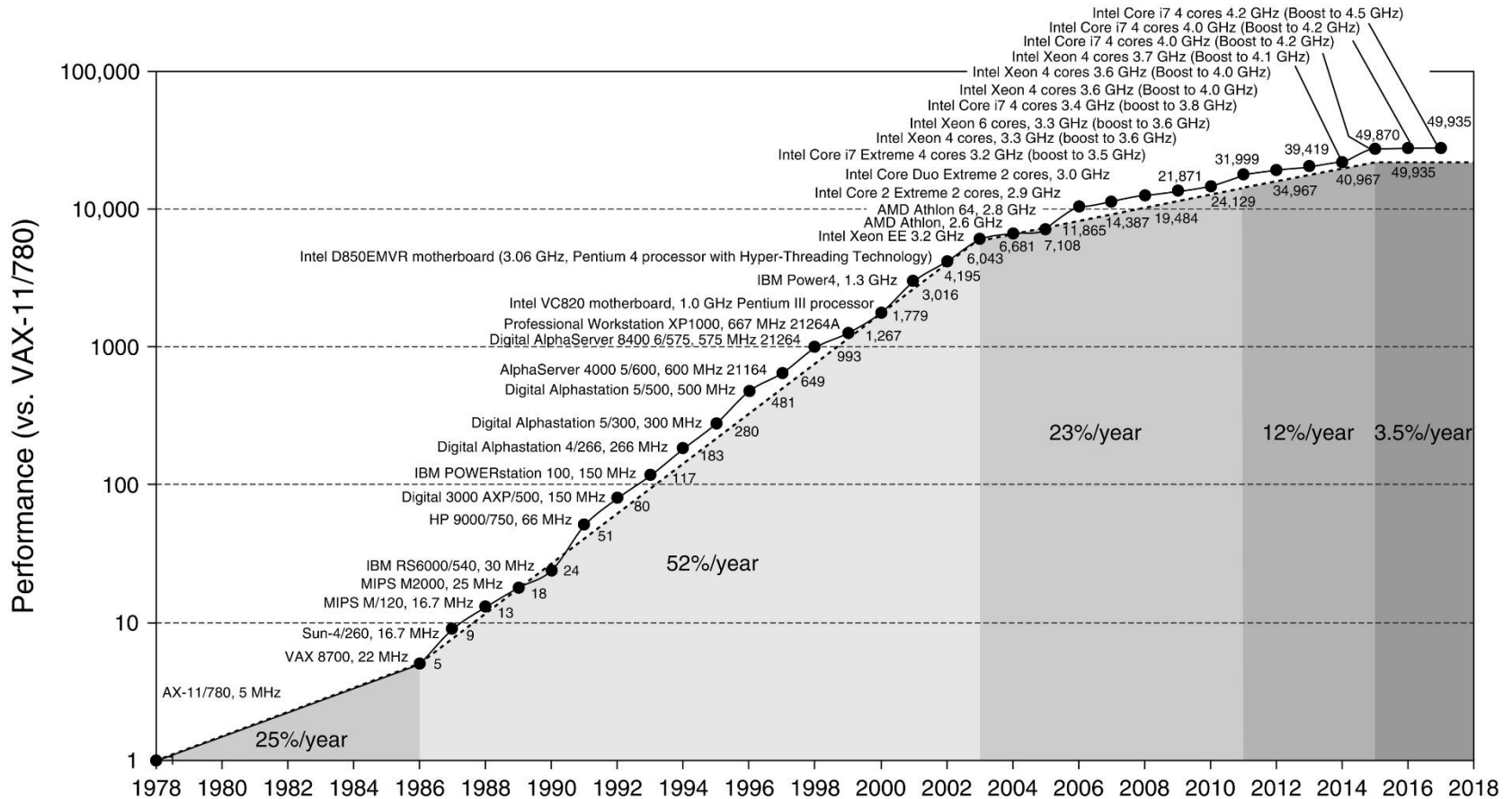
Dr. Gordon E. Moore is one of

摩尔定律的时代



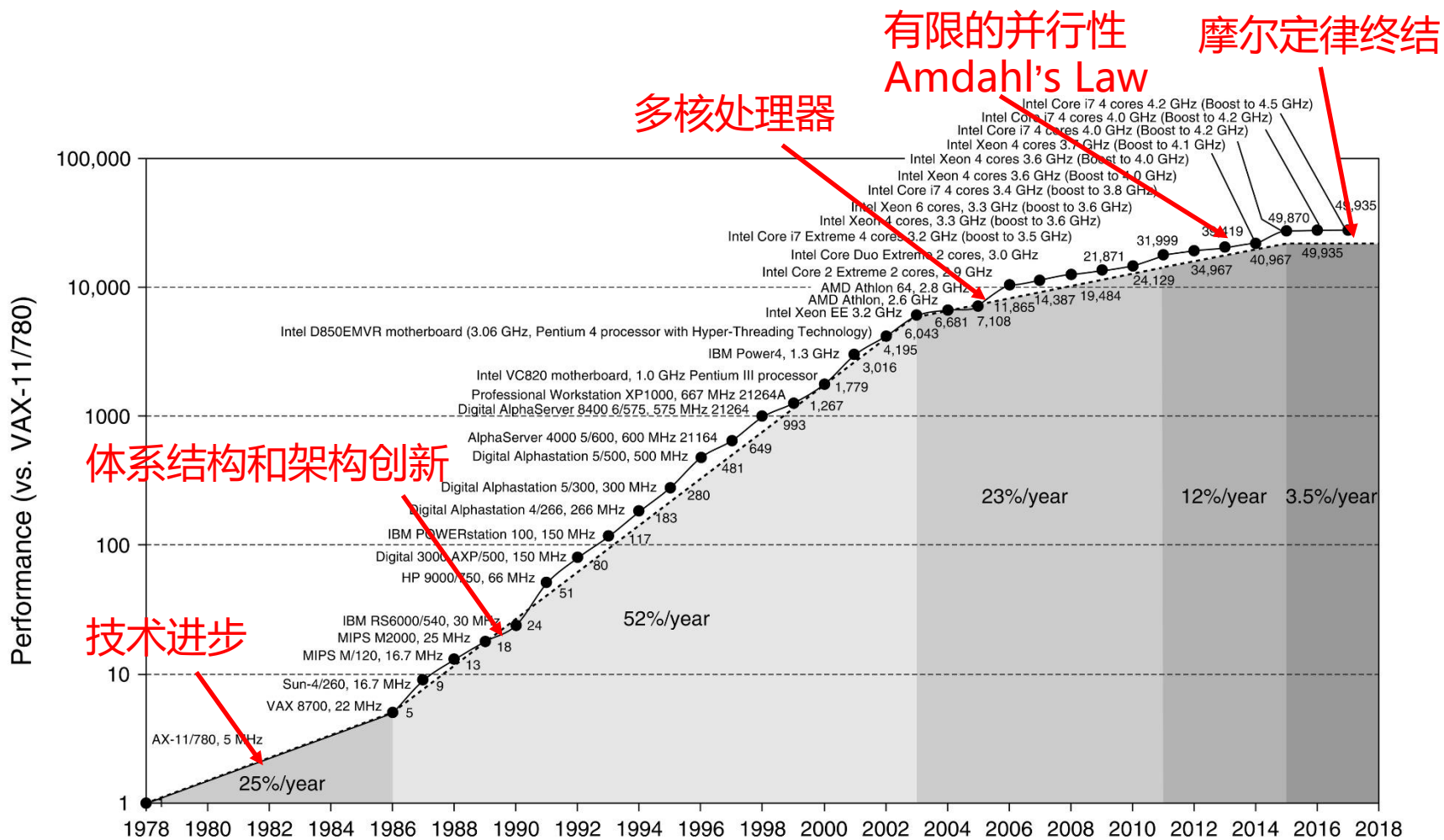
<http://www.intel.com/research/silicon/mooreslaw.htm>

摩尔定律的另一个视角



From Hennessy and Patterson 6e Image Copyright © 2019, Elsevier Inc. All rights Reserved.

各个时期的驱动力



From Hennessy and Patterson 6e Image Copyright © 2019, Elsevier Inc. All rights Reserved.

智能基座产教融合协同育人

华为联合72所高校共建 “智能基座产教融合协同育人基地”



教育部-华为签署战略合作备忘录



合肥工业大学-华为 “智能基座” 揭牌仪式



总结

- 什么是计算机体系结构？
- 抽象在CS中的重要性！
- CA课程的相关信息！！
- 计算机性能发展的趋势
- 功耗的组成部分和特性
- 作业：
 - 请阅读教材的第一章
 - 观看图灵奖演讲的视频